The estimator $\hat{\mu}_n$ is biased, since:

$$\mathbf{bias}(\hat{\mu}_n) \;=\; \mathrm{E}\left[\hat{\mu}_n\right] \;-\; \mu \;=\; \mathrm{E}\left[\alpha \sum_{i=1}^{n} (1 \;-\; \alpha)^{n-i}\, X_i\right] \;+\; (1 \;-\; \alpha)^n\, \mu_0 \;-\; \mu \tag{A169}$$

$$\begin{aligned}
&= \alpha \sum_{i=1}^{n} (1 \;-\; \alpha)^{n-i} \mathrm{E}\left[X_i\right] \;+\; (1 \;-\; \alpha)^n\, \mu_0 \;-\; \mu \\
&= \mu\, \alpha \sum_{i=0}^{n-1} (1 \;-\; \alpha)^{i} \;+\; (1 \;-\; \alpha)^n\, \mu_0 \;-\; \mu \\
&= \mu\, \left(1 \;-\; (1 \;-\; \alpha)^n\right) \;+\; (1 \;-\; \alpha)^n\, \mu_0 \;-\; \mu \;=\; (1 \;-\; \alpha)^n\, (\mu_0 \;-\; \mu)\,.
\end{aligned}$$

Asymptotically ($n \to \infty$) the estimate is unbiased. The variance is

$$\mathbf{var}(\hat{\mu}_n) \;=\; \mathrm{E}\left[\hat{\mu}_n^2\right] \;-\; \mathrm{E}^2\left[\hat{\mu}_n\right] \tag{A170}$$

$$\begin{aligned}
&= \mathrm{E}\left[\alpha^2 \sum_{i=1}^{n} \sum_{j=1}^{n} (1 \;-\; \alpha)^{n-i}\, X_i\, (1 \;-\; \alpha)^{n-j}\, X_j\right] \\
&+ \mathrm{E}\left[2\, (1 \;-\; \alpha)^n\, \mu_0\, \alpha \sum_{i=1}^{n} (1 \;-\; \alpha)^{n-i}\, X_i\right] \;+\; (1 \;-\; \alpha)^{2n}\, \mu_0^2 \\
&- \left((1 \;-\; \alpha)^n\, (\mu_0 \;-\; \mu) \;+\; \mu\right)^2 \\
&= \alpha^2\, \mathrm{E}\left[\sum_{i=1}^{n} (1 \;-\; \alpha)^{2(n-i)}\, X_i^2 \;+\; \sum_{i=1}^{n} \sum_{j=1, j\neq i}^{n} (1 \;-\; \alpha)^{n-i}\, X_i\, (1 \;-\; \alpha)^{n-j}\, X_j\right] \\
&+ 2\, (1 \;-\; \alpha)^n\, \mu_0\, \mu\, \alpha \sum_{i=1}^{n} (1 \;-\; \alpha)^{n-i} \;+\; (1 \;-\; \alpha)^{2n}\, \mu_0^2 \\
&- \left((1 \;-\; \alpha)^n\, \mu_0 \;+\; (1 \;-\; (1 \;-\; \alpha)^n)\, \mu\right)^2 \\
&= \alpha^2 \left(\sum_{i=1}^{n} (1 \;-\; \alpha)^{2(n-i)} \left(\sigma^2 \;+\; \mu^2\right) + \sum_{i=1}^{n} \sum_{j=1, j\neq i}^{n} (1 \;-\; \alpha)^{n-i}\, (1 \;-\; \alpha)^{n-j}\, \mu^2\right) \\
&+ 2\, (1 \;-\; \alpha)^n\, \mu_0\, \mu\, (1 \;-\; (1 \;-\; \alpha)^n) \;+\; (1 \;-\; \alpha)^{2n}\, \mu_0^2 \\
&- (1 \;-\; \alpha)^{2n}\, \mu_0^2 \;-\; 2\, (1 \;-\; \alpha)^n\, \mu_0\, (1 \;-\; (1 \;-\; \alpha)^n)\, \mu \;-\; (1 \;-\; (1 \;-\; \alpha)^n)^2\, \mu^2 \\
&= \sigma^2\, \alpha^2 \sum_{i=0}^{n-1} \left((1 \;-\; \alpha)^2\right)^i \;+\; \mu^2\, \alpha^2 \left(\sum_{i=0}^{n-1} (1 \;-\; \alpha)^i\right)^2 \;-\; (1 \;-\; (1 \;-\; \alpha)^n)^2\, \mu^2 \\
&= \sigma^2\, \alpha^2\, \frac{1 - (1 \;-\; \alpha)^{2n}}{1 - (1 \;-\; \alpha)^2} \;=\; \sigma^2\, \frac{\alpha\, (1 \;-\; (1 \;-\; \alpha)^{2n})}{2 - \alpha}\,.
\end{aligned}$$

Also the estimation variance of the exponentially weighted arithmetic mean is proportional to σ^2, which is the variance of the distribution the samples are drawn from.
The deviation of random variable X from its mean μ can be analyzed with Chebyshev's inequality. Chebyshev's inequality [15, 131] states that for a random variable X with expected value μ and variance $\tilde{\sigma}^2$ and for any real number $\epsilon > 0$:

$$\Pr\left[|X - \mu| \;\geqslant\; \epsilon\, \tilde{\sigma}\right] \;\leqslant\; \frac{1}{\epsilon^2} \tag{A171}$$

or, equivalently,

$$\Pr\left[|X - \mu| \;\geqslant\; \epsilon\right] \;\leqslant\; \frac{\tilde{\sigma}^2}{\epsilon^2}\,. \tag{A172}$$

For n samples $\{X_1, \ldots, X_n\}$ from a distribution with expectation μ and variance σ we compute the arithmetic mean $\frac{1}{n}\sum_{i=1}^{n} X_i$. If X is the arithmetic mean, then $\tilde{\sigma}^2 = \sigma^2/n$ and we obtain

$$\Pr\left[\left|\frac{1}{n}\sum_{i=1}^{n} X_i \;-\; \mu\right| \;\geqslant\; \epsilon\right] \;\leqslant\; \frac{\sigma^2}{n\,\epsilon^2}\;. \tag{A173}$$

Following Grünewälder and Obermayer [41], Bernstein's inequality can be used to describe the deviation of the arithmetic mean (unbiased estimator of μ) from the expectation μ (see Theorem 6 of Gábor Lugosi's lecture notes [75]):

$$\Pr\left[\left|\frac{1}{n}\sum_{i=1}^{n} X_i \;-\; \mu\right| \;\geqslant\; \epsilon\right] \;\leqslant\; 2\,\exp\left(-\,\frac{\epsilon^2\,n}{2\,\sigma^2 \;+\; \frac{2\,M\,\epsilon}{3}}\right)\;, \tag{A174}$$

where $|X - \mu| < M$.

A3.2 Mean and Variance of an MDP Sample of the Return

Since the variance of the estimators of the expectations (arithmetic mean and exponentially weighted arithmetic mean) is governed by the variance of the samples, we compute mean and variance of the return estimate $q^\pi(s,a)$. We follow [121, 129, 130] for deriving the mean and variance.
We consider an MDP with finite horizon T, that is, each episode has length T. The finite horizon MDP can be generalized to an MDP with absorbing (terminal) state $s = \mathtt{E}$. We only consider proper policies, that is there exists an integer n such that from any initial state the probability of achieving the terminal state $\mathtt{E}$ after n steps is strictly positive. T is the time to the first visit of the terminal state: $T = \min k \mid s_k = \mathtt{E}$. The return G_0 is:

$$G_0 \;=\; \sum_{k=0}^{T} \gamma^k\, R_{k+1}\;. \tag{A175}$$

The action-value function, the Q-function, is the expected return

$$G_t \;=\; \sum_{k=0}^{T-t} \gamma^k\, R_{t+k+1} \tag{A176}$$

if starting in state $S_t = s$ and action $A_t = a$:

$$q^\pi(s,a) \;=\; \mathrm{E}_\pi\left[G_t \mid s,a\right]\;. \tag{A177}$$

The second moment of the return is:

$$M^\pi(s,a) \;=\; \mathrm{E}_\pi\left[G_t^2 \mid s,a\right]\;. \tag{A178}$$

The variance of the return is:

$$V^\pi(s,a) \;=\; \mathrm{Var}_\pi\left[G_t \mid s,a\right] \;=\; M^\pi(s,a) \;-\; \left(q^\pi(s,a)\right)^2\;. \tag{A179}$$

Using $\mathrm{E}_{s',a'}(f(s',a')) = \sum_{s'} p(s' \mid s,a) \sum_{a'} \pi(a' \mid s') f(s',a')$, and analogously $\mathrm{Var}_{s',a'}$ and Var_r, the next Theorem A7 gives mean and variance $V^\pi(s,a) = \mathrm{Var}_\pi\left[G_t \mid s,a\right]$ of sampling returns from an MDP.

Theorem A7. *The mean q^π and variance V^π of sampled returns from an MDP are*

$$q^\pi(s,a) = \sum_{s',r} p(s',r \mid s,a)\left(r + \gamma \sum_{a'} \pi(a' \mid s') q^\pi(s',a')\right) = r(s,a) + \gamma \mathrm{E}_{s',a'}\left[q^\pi(s',a') \mid s,a\right],$$

$$V^\pi(s,a) = \mathrm{Var}_r\left[r \mid s,a\right] + \gamma^2\left(\mathrm{E}_{s',a'}\left[V^\pi(s',a') \mid s,a\right] + \mathrm{Var}_{s',a'}\left[q^\pi(s',a') \mid s,a\right]\right). \tag{A180}$$

Proof. The Bellman equation for Q-values is

$$\begin{aligned} q^\pi(s,a) \;&=\; \sum_{s',r} p(s',r \mid s,a)\left(r \;+\; \gamma \sum_{a'} \pi(a' \mid s')\, q^\pi(s',a')\right) \\ &=\; r(s,a) \;+\; \gamma\, \mathrm{E}_{s',a'}\left[q^\pi(s',a') \mid s,a\right]\;. \end{aligned} \tag{A181}$$

This equation gives the mean if drawing one sample. We use

$$r(s,a) \;=\; \sum_r r\; p(r \mid s,a)\;, \tag{A182}$$

$$r^2(s,a) \;=\; \sum_r r^2\; p(r \mid s,a)\;. \tag{A183}$$

For the second moment, we obtain [129]:

$$\begin{aligned}
M^\pi(s,a) \;&=\; \mathrm{E}_\pi\left[G_t^2 \mid s,a\right] \qquad \text{(A184)}\\
&=\; \mathrm{E}_\pi\left[\left(\sum_{k=0}^{T-t}\gamma^k R_{t+k+1}\right)^2 \mid s,a\right]\\
&=\; \mathrm{E}_\pi\left[\left(R_{t+1} \;+\; \sum_{k=1}^{T-t}\gamma^k\; R_{t+k+1}\right)^2 \mid s,a\right]\\
&=\; r^2(s,a) \;+\; 2\, r(s,a)\, \mathrm{E}_\pi\left[\sum_{k=1}^{T-t}\gamma^k\; R_{t+k+1} \mid s,a\right]\\
&+\; \mathrm{E}_\pi\left[\left(\sum_{k=1}^{T-t}\gamma^k\; R_{t+k+1}\right)^2 \mid s,a\right]\\
&=\; r^2(s,a) \;+\; 2\gamma\, r(s,a) \sum_{s'} p(s' \mid s,a) \sum_{a'} \pi(a' \mid s')\; q^\pi(s',a')\\
&+\; \gamma^2 \sum_{s'} p(s' \mid s,a) \sum_{a'} \pi(a' \mid s')\; M^\pi(s',a')\\
&=\; r^2(s,a) \;+\; 2\gamma\, r(s,a)\, \mathrm{E}_{s',a'}\left[q^\pi(s',a') \mid s,a\right] \;+\; \gamma^2\, \mathrm{E}_{s',a'}\left[M^\pi(s',a') \mid s,a\right]\;.
\end{aligned}$$

For the variance, we obtain:

$$\begin{aligned}
V^\pi(s,a) \;&=\; M^\pi(s,a) \;-\; \left(q^\pi(s,a)\right)^2 \qquad \text{(A185)}\\
&=\; r^2(s,a) \;-\; (r(s,a))^2 \;+\; \gamma^2\, \mathrm{E}_{s',a'}\left[M^\pi(s',a') \mid s,a\right] \;-\; \gamma^2\, \mathrm{E}^2_{s',a'}\left[q^\pi(s',a') \mid s,a\right]\\
&=\; \mathrm{Var}_r\left[r \mid s,a\right] \;+\; \gamma^2\; \Big(\mathrm{E}_{s',a'}\left[M^\pi(s',a') \;-\; \left(q^\pi(s',a')\right)^2 \mid s,a\right]\\
&-\; \mathrm{E}^2_{s',a'}\left[q^\pi(s',a') \mid s,a\right] \;+\; \mathrm{E}_{s',a'}\left[\left(q^\pi(s',a')\right)^2 \mid s,a\right]\Big)\\
&=\; \mathrm{Var}_r\left[r \mid s,a\right] \;+\; \gamma^2\; \left(\mathrm{E}_{s',a'}\left[V^\pi(s',a') \mid s,a\right] \;+\; \mathrm{Var}_{s',a'}\left[q^\pi(s',a') \mid s,a\right]\right)\;.
\end{aligned}$$

□

For deterministic reward, that is, $\mathrm{Var}_r\left[r \mid s,a\right] = 0$, the corresponding result is given as Equation (4) in Sobel 1982 [121] and as Proposition 3.1 (c) in Tamar et al. 2012 [129].
For temporal difference (TD) learning, the next Q-values are fixed to $\hat{q}^\pi(s',a')$ when drawing a sample. Therefore, TD is biased, that is, both SARSA and Q-learning are biased. During learning with according updates of Q-values, $\hat{q}^\pi(s',a')$ approaches $q^\pi(s',a')$, and the bias is reduced. However, this reduction of the bias is exponentially small in the number of time steps between reward and updated Q-values, as we will see later. The reduction of the bias is exponentially small for eligibility traces, too.
The variance recursion Eq. (A180) of sampled returns consists of three parts:

- (1) the immediate variance $\mathrm{Var}_r\left[r \mid s,a\right]$ of the immediate reward stemming from the probabilistic reward $p(r \mid s,a)$,
- (2) the local variance $\gamma^2 \mathrm{Var}_{s',a'}\left[q^\pi(s',a') \mid s,a\right]$ from state transitions $p(s' \mid s,a)$ and new actions $\pi(a' \mid s')$,
- (3) the expected variance $\gamma^2 \mathrm{E}_{s',a'}\left[V^\pi(s',a') \mid s,a\right]$ of the next Q-values.

For different settings the following parts may be zero: